1 Fair Division

1.1 Methods by Name

Divider-Chooser, Lone Divider, the Method of Sealed Bids.

1.2 Sample Fair Divison Problems

1. For each of the three methods listed above, what is the **context** in which they are appropriate? (for example: how many people are dividing? How can you split up the stuff?)

2. The Cookie Monster and Elmo are splitting a giant cookie that is half chocolate-chip and half oatmeal, worth $12. The Cookie Monster likes chocolate chip twice as much as oatmeal. Elmo likes oatmeal three times as much as chocolate chip.

   (a) Suppose someone splits the cookie with all chocolate chip on one side and all oatmeal on the other. Attach dollar values to each share for each person.

	share 1: chocolate chip	share 2: oatmeal
Cookie Monster		
Elmo		

   (b) Which cookie pieces (if any) are fair shares for Elmo? Which are fair shares for Cookie Monster? Show calculations that support your answer.

3. (a) James, Gordon, Julie, Alexei, and Latrice divide their pile of Halloween candy worth a total of \$20. Determine who was the divider and determine how the lone divider method would proceed.

	Pile 1	Pile 2	Pile 3	Pile 4	Pile 5
James	\$0	\$5	\$10	\$5	\$0
Gordon	\$2	\$4	\$6	\$2	\$6
Julie	\$4	\$4	\$4	\$4	\$4
Alexei	\$2	\$4	\$12	\$2	\$0
Latrice	\$2	\$10	\$7	\$1	\$0

(b) Suppose **Gordon** changes his evaluation of the piles. Now determine how the lone divider method would proceed.

	Pile 1	Pile 2	Pile 3	Pile 4	Pile 5
James	\$0	\$5	\$10	\$5	\$0
Gordon	\$2	**\$7**	\$6	\$2	**\$3**
Julie	\$4	\$4	\$4	\$4	\$4
Alexei	\$2	\$4	\$12	\$2	\$0
Latrice	\$2	\$10	\$7	\$1	\$0

4. Harry Potter (P), Hermione Granger (G), Luna Lovegood (L), and Ronald Weasley (W) are dividing up some loot they found in the dungeons. The loot consists of a velvet cloak, a gold chalice, and a self-cleaning cauldron. They decide to divide the loot using the sealed bid method. The table below shows how many gold coins each person bid for each item.

   (a) Fill in the following table.

	Potter	Granger	Lovegood	Weasley
cloak	60 coins	50 coins	100 coins	30 coins
chalice	60 coins	10 coins	0 coins	20 coins
cauldron	0 coins	80 coins	100 coins	30 coins
total bid				
What is a **fair share** for each person?				
award (who gets what item)				
award value				
How much money does each person **pay in** or **receive** from the holding pile?				
Determine the surplus.				
Determine each person's fair share of the surplus.				

   (b) Determine the Final Allocation by filling out each blank below. Space is provided here to determine the total cash for each person.

Summary of Final Allocation			
person	items	cash	paid or received?
Pottter			
Granger			
Lovegood			
Weasley			

   (c) Is this a fair allocation? Write a computation to explain how you know.

2 Graph Theory

1. Define the terms below.

   (a) An **Euler circuit** is

   (b) An **Euler path** is

   (c) A **Hamiltonian circuit** is

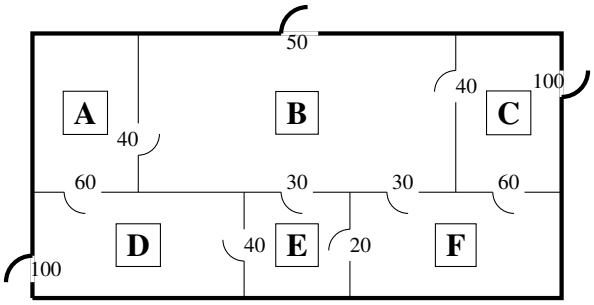
   (d) A **spanning tree** is

   (e) A **minimum cost spanning tree** is

   (f) the **degree of a vertex** is

   (g) A **complete graph** is

2. Below is the floor plan of an art gallery with 6 galleries and 11 doorways. A security company has assigned a theft risk assessment for each doorway. The higher the number, the greater the risk. Each gallery is labeled with a letter.



(a) Draw a graph below that models this scenario.

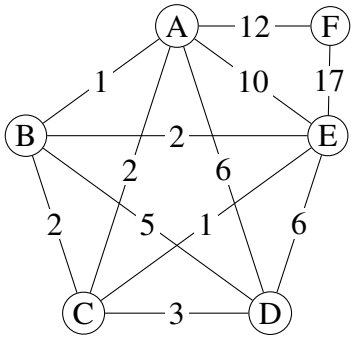
(b) The gallery owners want all rooms to be available to visitors but they would like to shut and lock as many doors as possible to lower their risk. **Using terminology from Graph Theory**, what is the company looking for?

3. (a) Given the following table of distances (in miles) between towns in Alaska, construct a weighted graph model that represents the situation. A – in the table means there is not a direct route between those two towns (or there is a route but it passes through some other town on the list).

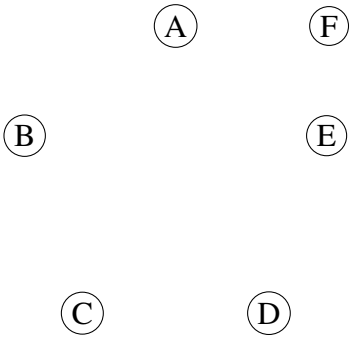
	Anchorage	Cantwell	Delta Junction	Fairbanks	Glennallen	Palmer	Tok
Anchorage	–	210	–	–	–	42	–
Cantwell	210	–	–	148	–	–	–
Delta Junction	–	–	–	98	151	–	108
Fairbanks	–	148	98	–	–	–	–
Glennallen	–	–	151	–	–	147	139
Palmer	42	–	–	–	147	–	–
Tok	–	–	108	–	139	–	–

(b) Is the graph connected? Why or why not? Use terminology from graph theory to explain your answer.

4. (a) Use **Kruskal's Algorithm** to find a **minimum cost spanning tree** in the graph below. Document each edge you looked at, in order, and whether or not that edge is used in the MCST.



edge	weight	used?

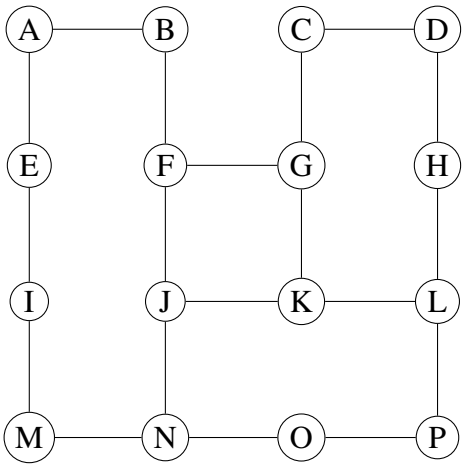


(b) Give an example of a real-world problem where you would want to find a minimum weight spanning tree. (You need to explain what the vertices represent, what the edges represent, and what the weights represent, and how a minimum weight spanning tree is useful.)

5. (a) What does it mean to **Eulerize** a graph? What is your goal? What kind of edges can you add?

   (b) **Eulerize** the graph below by duplicating the fewest number of edges and then find an **Euler circuit** in the resulting graph. Make sure that someone else can clearly see the additional edges. **List** the additional edges and the vertices in the order they are traversed in the Euler circuit.

   Offset your circuit slightly from the graph so that it is clear where the circuit is, and draw it as one **continuous** path. If possible use a different color.

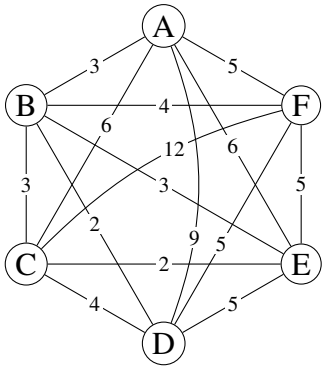


   added edges: _____

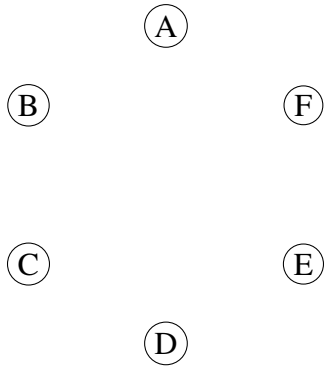
   Euler Circuit: _____

   (c) Give an example of a real-world problem where you would want to find an Euler circuit. (You need to explain what the vertices represent, what the edges represent, what the weights represent, and why finding an Euler circuit is useful in answering the problem.)

6. (a) Use the **Nearest Neighbor Algorithm** starting a vertex A, to find a **Hamiltonian Circuit** in the graph below and determine its weight. List the edges in the table in the order you considered them / added them.



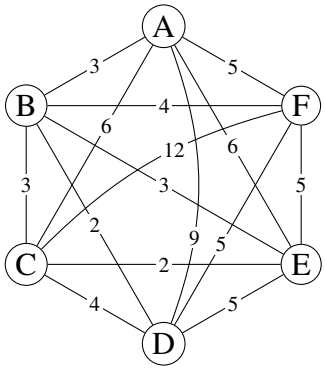
edge	weight	used?



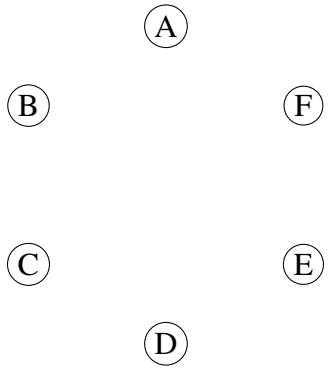
circuit: _____

weight: _____

(b) Use the **Cheapest Link Algorithm** to find a **Hamiltonian Circuit** in the graph below, and determine its weight. List the edges in the table in the order you considered them / added them.



edge	weight	used?



circuit: _____

weight: _____

(c) Give an example of a real-world problem where you would want to find a minimum weight Hamiltonian circuit. (You need to explain what the vertices represent, what the edges represent, and what the weights represent, and why finding a Hamiltonian circuit is useful in answering the problem.)